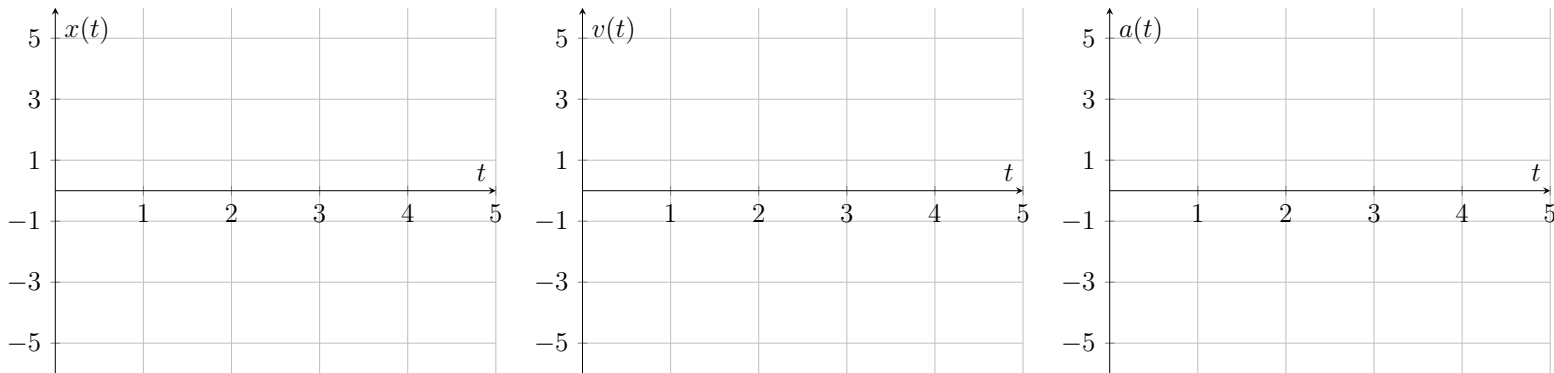


3. Below, plot the function $x(t) = 2x^{\frac{1}{2}}$, plot this as well as $v(t)$ and $a(t)$ of the same function



4. A large motor is running at 4000 rpm with an internal linear velocity of $v = 20 \text{ m s}^{-1}$. What is the radius R of the motor and the frequency f at which the motor is rotating?
5. Enzo is driving a remote control car on a train that has just begun to move after making a stop at a station
- (a) If the train is accelerating at 2 m s^{-2} at $t = 0 \text{ s}$ and Enzo's toy car is going at a constant speed of -3 m s^{-1} , what is the toy car's position as a function of time as seen from the stationary platform of the train station?
- (b) At what point will the toy car appear to be moving in the same direction as the train?